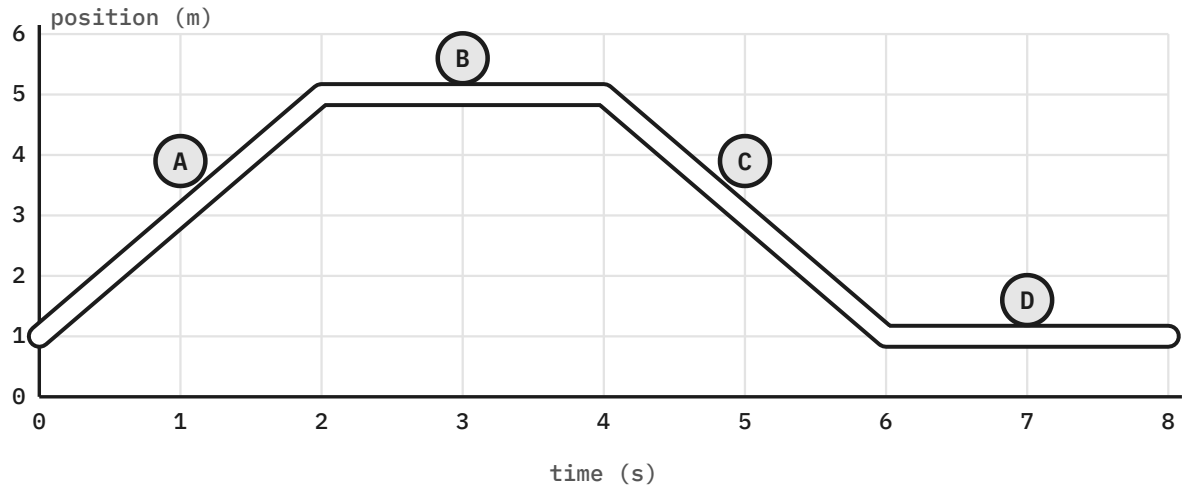
Read every part of the x–t graph

03 F

DIRECTIONS · Match each segment to its card by letter, then trace it purple: its slope is the velocity. Height = position. Slope = velocity.



A · 0–2 s
v = ____ m/s
away / back / stopped

B · 2–4 s
v = ____ m/s
away / back / stopped

C · 4–6 s
v = ____ m/s
away / back / stopped

D · 6–8 s
v = ____ m/s
away / back / stopped

SPEED OR VELOCITY? · WHOLE TRIP 0–8 s

distance ____ m · displacement ____ m
average speed = ____ m/s · average velocity = ____ m/s

TELL THE MOTION STORY

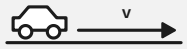
First the walker..., then..., finally...

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: velocity

Language goal: I can **explain** what slope means.



+5 m/s →

velocity v

velocidad · velocidade

how fast + which way

my language: _____

use it: _____



5 m/s · no direction

5 m/s

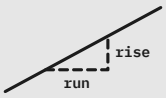
speed

rapidez · rapidez

how fast only

my language: _____

use it: _____



rise ÷ run

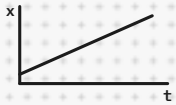
slope

pendiente · inclinação

how steep a line is

my language: _____

use it: _____



x ↑
→ t

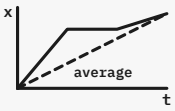
x–t graph

gráfica x–t · gráfico x–t

position on the up axis

my language: _____

use it: _____



total ÷ time

average

promedio · média

one value for a whole trip

my language: _____

use it: _____



v = 0

v = 0

at rest

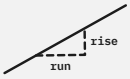
en reposo · em repouso

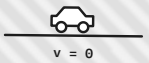
not moving

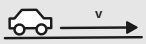
my language: _____

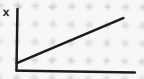
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner □

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Hallway sprint – the replay

● LIVE

You're the broadcast analyst. Practice data, not a real race.

03 X

SPLIT DATA

t (s)	x (m)	Δx (m)	v (m/s)
0	0	–	–
1	2		
2	6		
3	10		
4	14		
5	16		

SLOW-MO x–t GRAPH

STAT 1 · FASTEST SECOND

Which? ____

How does the graph show it?

STAT 2 · AVERAGE v, 0–5 s

v = ____ m/s

STAT 3 · STEADY STRETCH

When was v constant? ____

COMMENTARY BOOTH

"Off the line she covers ____ m in the first second… by second 3 she's holding a steady ____ m/s – you can see it in the ____ of the line!"

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every problem

03 M

DIRECTIONS · Work each problem through all four boxes, in order.

PROBLEM 1 · A walker moves +12 m in 4 s. What is the average velocity?

1 READ

Δx = ____ · Δt = ____ · find ____

2 DRAW

x–t sketch

3 SOLVE

$v = \Delta x \div \Delta t$

4 WRITE

The average velocity is ____ m/s, meaning ____ .

PROBLEM 2 · A student walks 8 m out and 8 m back in 10 s. Find average speed and average velocity.

1 READ

distance ____ · displacement ____ · time ____

2 DRAW

route sketch

3 SOLVE

speed = distance ÷ t · $v = \Delta x \div t$

4 WRITE

Speed is ____ but velocity is ____ because ____ .

Hardest box: □ Read □ Draw □ Solve □ Write

S1 1 2 3

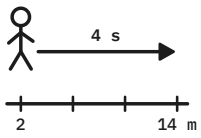
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Talk it through: velocity

Language goal: I can **explain** a graph's slope.

DIRECTIONS · Partner A and B read their lines aloud. Listen, then write. Switch each round.

ROUND 1 · A WALKER: 2 m → 14 m IN 4 s

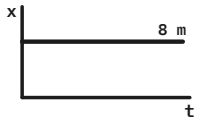


A ASKS "What is the walker's velocity?"

B SAYS "The change in position is ____ m. I divide by ____ s. So v = ____ m/s."

A CHECKS "So you're saying the walker moves ____ m every second. I agree / disagree because ____."

ROUND 2 · SWITCH · A FLAT x–t LINE AT 8 m

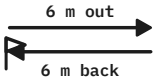


B ASKS "Is the walker fast because the line is high?"

A SAYS "No. The height shows ____. The slope is ____, so the walker is ____."

B ADDS "I want to add that velocity comes from the ____ of the line."

ROUND 3 · BOTH · A ROUND TRIP, 6 m OUT AND BACK IN 4 s



A "The average speed is ____ because the distance is ____."

B "But the average velocity is ____ because the displacement is ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

Velocity is different from speed because…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: velocity

03 C

DIRECTIONS · Solve each station. Copy the letter beside your answer into the lock.

STATION 1

x goes from 2 m to 14 m in 4 s. Find v.

(S) +3 m/s (K) +3.5 m/s (V) 12 m/s

STATION 2

x goes from 9 m to 3 m in 2 s. Find v.

(A) +3 m/s (L) −3 m/s (U) −6 m/s

STATION 3

An x–t line is flat at x = 8 m. Find v.

(O) 0 m/s (P) 8 m/s

STATION 4

6 m east, then 6 m west, in 4 s. Average speed?

(P) 3 m/s (N) 0 m/s

STATION 5

Same trip. Average velocity?

(E) 0 m/s (R) 3 m/s

STATION 6

8,500 m/s in km/h (× 3.6)?

(S) 30,600 km/h (T) 2,361 km/h

LOCK [1] [2] [3] [4] [5] [6]

Hint: what we call the steepness of x–t graphs (plural).

S1 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: same speed, different velocity

03 CS

YOUR READER
A 5th grader who thinks speed and velocity are the same thing.
Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ PURPLE — A velocity arrow on each runner, with + or – written on it
- ☐ ◌ BLUE — Position marks on the number line with numbers
- ☐ ◯ PENCIL — The speed written as **4 m/s** for both

1 · SAME SPEED *Two runners both moving at 4 m/s. Put them on a number line.*

CAPTION _____

2 · OPPOSITE WAYS *Show that they head in opposite directions.*

CAPTION _____

3 · THE DIFFERENCE *A character explains why their velocities are not the same.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **speed · velocity · direction · position**

_____

_____

S1 1 2 3

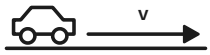
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · velocity

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Velocity · velocidad (ES) · velocidade (PT)



TODAY'S CASE

One cart goes from 8 m to 2 m in 3 s. Another goes from 2 m to 8 m in 3 s.

01 · SKETCH AND LABEL

Sketch both position–time lines on the same axes. Label t (s) and x (m).

02 · CHANGE THE PICTURE

Draw a purple velocity arrow for each cart. Use the two slopes to decide what is equal and what has opposite signs.

Annotate your sketches. Circle the mark that best shows velocity. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

Both carts have the same speed, but …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · velocity

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Velocity · velocidad (ES) · velocidade (PT)

A · The cart's speed is 2 m/s, with no direction given.

B · A cart moves at 2 m/s to the left.

01 · DEFINITION

What does velocity mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else